

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce **Department of Information Technology (B.Sc.I.T Semester V)**
Applied Business Analytics

Practical – 1

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Class: TYIT	Batch: 1
Date of Assignment: 24/07/26	Date/Time of Submission: 24/07/26

Aim:

a. Collect data from a real-life business scenario and perform exploratory data analysis (EDA) to gain insights into the dataset.

Code & Output:

Step 1: imports all necessary files.

Step 2: load the data set and display the first & last 5 rows.

Step 3: display the shape of dataset and column name & information.

```
#Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

#Load dataset
df = pd.read_csv("Retail_sales.csv")

#Display First 5 Records
df.head()

#Display Last 5 Records
df.tail()

#Display Shape of Dataset
df.shape

#Display Column Names
df.columns

#Display Dataset Information
df.info()
```

	Store ID	Product ID	Date	Units Sold	Sales Revenue (USD)	Discount
0	Spearsland	52372247	2022-01-01	9	2741.69	
1	Spearsland	52372247	2022-01-02	7	2665.53	
2	Spearsland	52372247	2022-01-03	1	380.79	
3	Spearsland	52372247	2022-01-04	4	1523.16	
4	Spearsland	52372247	2022-01-05	2	761.58	

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 11 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   Store ID            30000 non-null  object
 1   Product ID          30000 non-null  int64
 2   Date                30000 non-null  object
 3   Units Sold          30000 non-null  int64
 4   Sales Revenue (USD) 30000 non-null  float64
 5   Discount Percentage  30000 non-null  int64
 6   Marketing Spend (USD) 30000 non-null  int64
 7   Store Location       30000 non-null  object
 8   Product Category    30000 non-null  object
 9   Day of the Week     30000 non-null  object
10   Holiday Effect       30000 non-null  bool
dtypes: bool(1), float64(1), int64(4), object(5)
memory usage: 2.3+ MB
```

Step 4: Display data type and check the missing value & duplicate records.

Step 5: Display dscrip static, count uniq value & display unique value of particular columns

```
#Display Data Types
df.dtypes

Store ID      object
Product ID    int64
Date          object
Units Sold    int64
Sales Revenue (USD) float64
Discount Percentage int64
Marketing Spend (USD) int64
Store Location object
Product Category object
Day of the Week object
Holiday Effect bool
dtype: object

#Check Missing Values
df.isnull().sum()

Store ID      0
Product ID    0
Date          0
Units Sold    0
Sales Revenue (USD) 0
Discount Percentage 0
Marketing Spend (USD) 0
Store Location 0
Product Category 0
Day of the Week 0
Holiday Effect 0
dtype: int64

#Check Duplicate Records
df.duplicated().sum()

np.int64(0)

#Display Descriptive Statistics
df.describe()

Product ID  Units Sold  Sales Revenue (USD)  Discount Percentage  Marketing Spend (USD)
count  3.000000e+04  30000.000000  30000.000000  30000.000000  30000.000000
mean    4.461294e+07   6.161967    2749.509593    2.973833    49.944033
std     2.779759e+07   3.323929    2568.639288    5.974530    64.401655
min     3.636541e+06   0.000000     0.000000     0.000000     0.000000
25%    2.228600e+07   4.000000     882.592500    0.000000     0.000000
50%    4.002449e+07   6.000000    1902.420000    0.000000     1.000000
75%    6.559352e+07   8.000000    3863.920000    0.000000    100.000000
max     9.628253e+07  56.000000   27165.880000   20.000000   199.000000

#Count Unique Values
df.nunique()

Store ID      1
Product ID    42
Date          731
Units Sold    32
Sales Revenue (USD) 2545
Discount Percentage 5
Marketing Spend (USD) 200
Store Location 243
Product Category 4
Day of the Week 7
Holiday Effect 2
dtype: int64

#Display Unique Values of a Particular Column
df["Product Category"].unique()

array(['Furniture', 'Electronics', 'Groceries', 'Clothing'], dtype=object)
```

Step 6: count product category and count store location, holliday effect.

Step 7: calculate sale revenue , maximum & minimum sale revenue , correlation matrix.

```
#Count Product Categories
df["Product Category"].value_counts()

Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64

#Count Store Locations
df["Store Location"].value_counts()

Store Location
Korea      237
Congo      234
Turks and Caicos Islands 156
Guam       153
Tanzania   151
...
Trinidad and Tobago    98
Haiti                  96
Dominica               95
Ireland                92
Andorra                80
Name: count, Length: 243, dtype: int64

#Count Holiday Effect
df["Holiday Effect"].value_counts()

Holiday Effect
False    29836
True      164
Name: count, dtype: int64

#Calculate Average Sales Revenue
df["Sales Revenue (USD)"].mean()

np.float64(2749.509592666667)

#Find Maximum Sales Revenue
df["Sales Revenue (USD)"].max()

np.float64(27165.88)

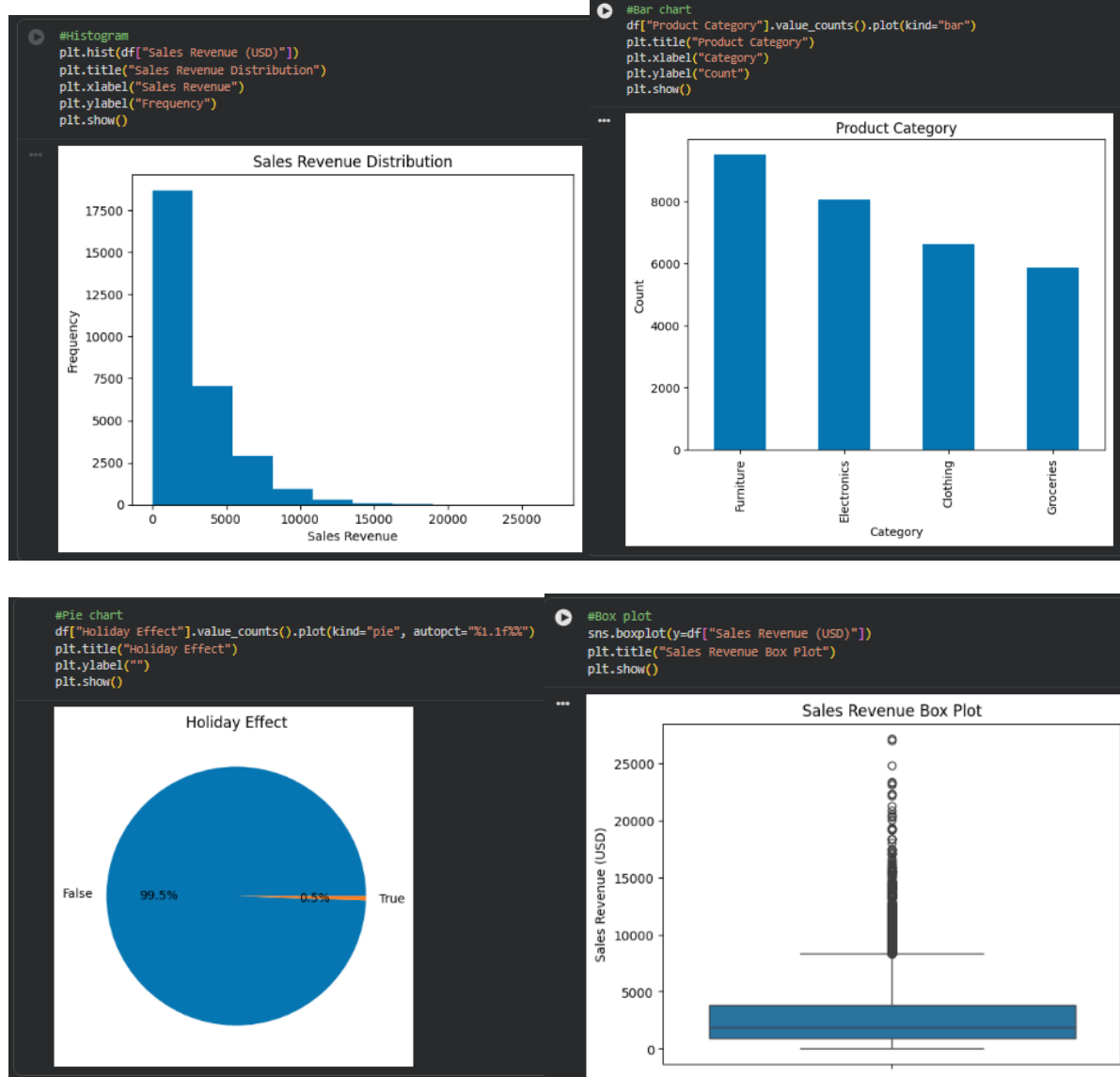
#Find Minimum Sales Revenue
df["Sales Revenue (USD)"].min()

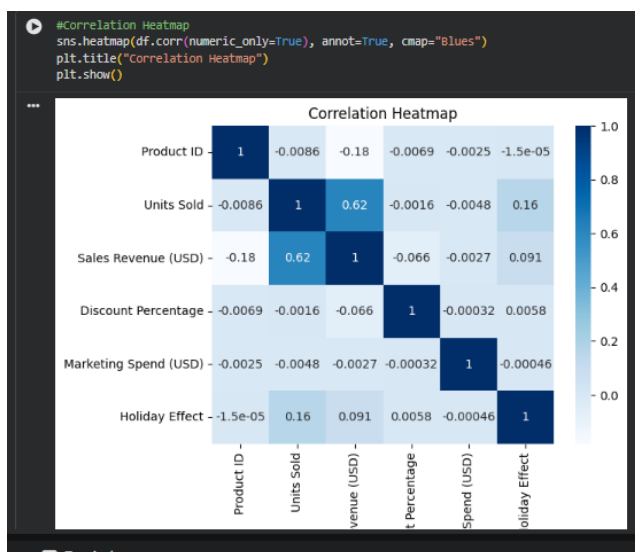
np.float64(0.0)

#Correlation Matrix
df.corr(numeric_only=True)

Product ID  Units Sold  Sales Revenue (USD)  Discount Percentage  Marketing Spend (USD)  Holiday Effect
Product ID    1.000000    -0.008570        -0.181934        -0.006891        -0.002496    -0.000015
Units Sold    -0.008570    1.000000         0.615767        -0.001620        -0.004750    0.159324
Sales Revenue (USD) -0.181934    0.615767         1.000000        -0.065791        -0.002732    0.090681
Discount Percentage -0.006891   -0.001620       -0.065791         1.000000        -0.000322    0.005848
Marketing Spend (USD) -0.002496   -0.004750       -0.002732       -0.000322         1.000000    -0.000455
Holiday Effect -0.000015    0.159324         0.090681         0.005848        -0.000455    1.000000
```

Step 8: Make a histogram , Bar , pie , Box and correlation heatmap.





b. Analyze customer data to identify trends and patterns that can be used for business decision-making.

Code & Output:

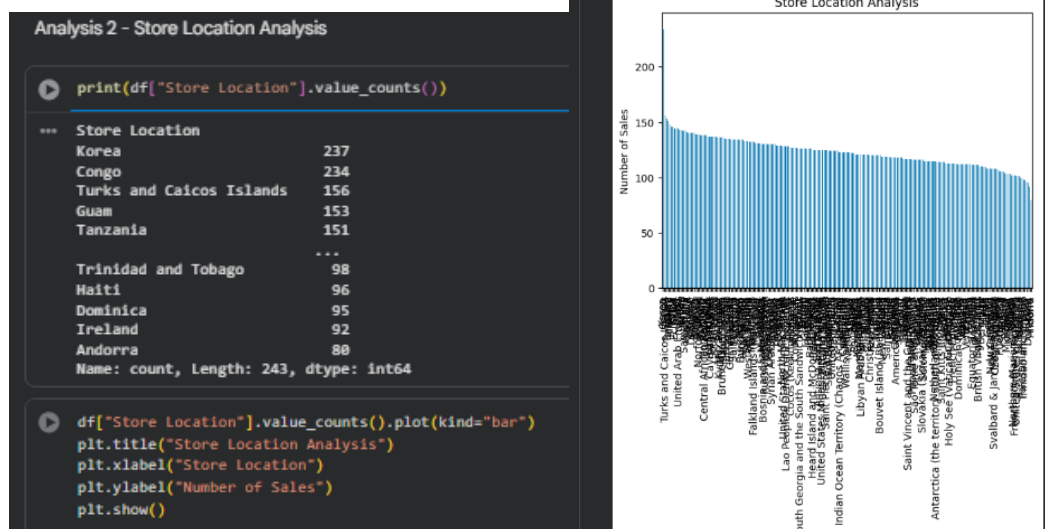
Step 1: Import Required Libraries

Step 2: load the dataset

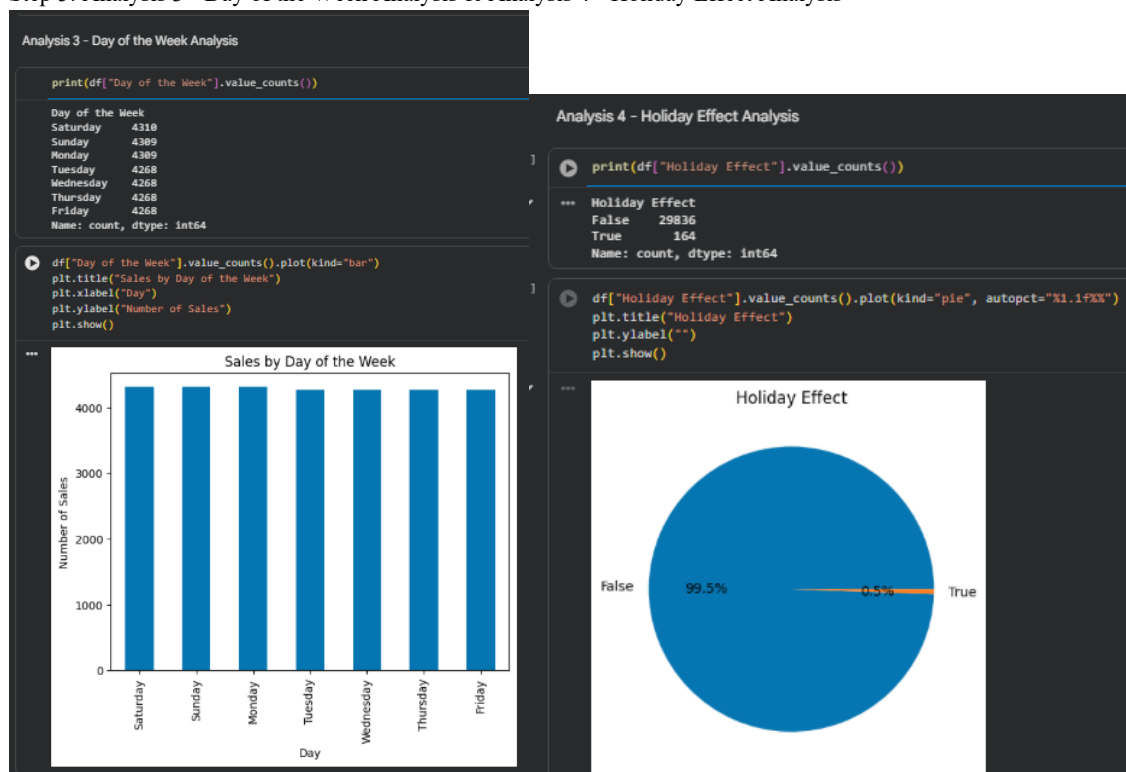
Step 3: Analysis 1 - Product category analysis



Step 4: Analysis 2 - Store Location Analysis



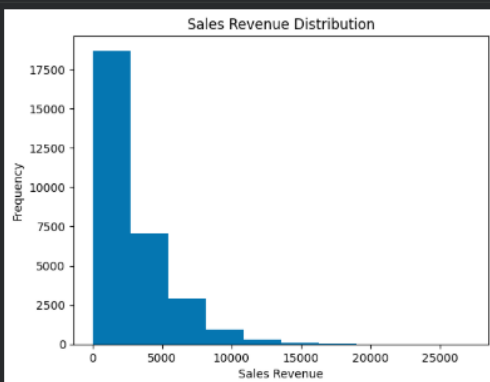
Step 5: Analysis 3 - Day of the Week Analysis & Analysis 4 - Holiday Effect Analysis



Step 6: Analysis 5 - Sales Revenue Analysis & Analysis 6 - Units Sold Analysis

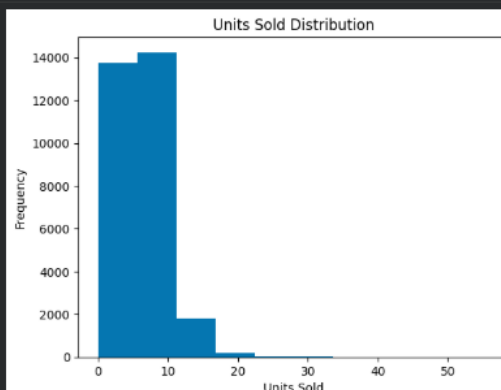
Analysis 5 - Sales Revenue Analysis

```
plt.hist(df["Sales Revenue (USD)"], bins=10)
plt.title("Sales Revenue Distribution")
plt.xlabel("Sales Revenue")
plt.ylabel("Frequency")
plt.show()
```



Analysis 6 - Units Sold Analysis

```
plt.hist(df["Units Sold"], bins=10)
plt.title("Units Sold Distribution")
plt.xlabel("Units Sold")
plt.ylabel("Frequency")
plt.show()
```



Step 7: Analysis 7 - Marketing Spend vs Sales Revenue & Analysis 8 - Correlation Analysis

Analysis 7 - Marketing Spend vs Sales Revenue

```
plt.scatter(df["Marketing Spend (USD)"], df["Sales Revenue (USD)"])
plt.title("Marketing Spend vs Sales Revenue")
plt.xlabel("Marketing Spend")
plt.ylabel("Sales Revenue")
plt.show()
```



Analysis 8 - Correlation Analysis

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Oranges")
plt.title("Correlation Heatmap")
plt.show()
```

